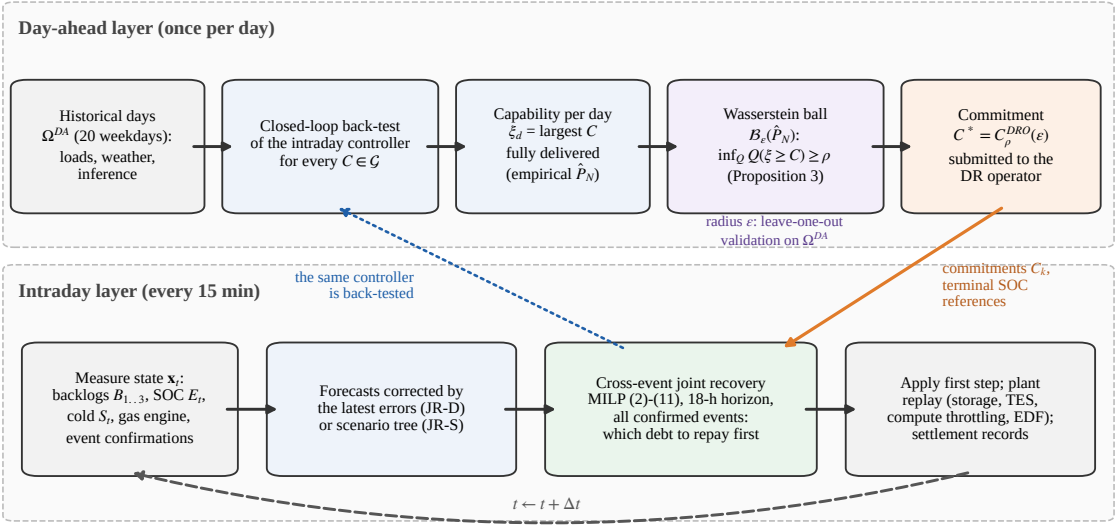
Day-ahead layer (once per day)
Historical days
Ω^{DA} (20 weekdays):
loads, weather,
inference
Closed-loop back-test
of the intraday controller
for every $C \in \mathcal{G}$
Capability per day
ξ_d = largest C
fully delivered
(empirical $\hat{P}_N$)
Wasserstein ball
$\mathcal{B}_\varepsilon(\hat{P}_N)$:
$\inf_Q Q(\xi \geq C) \geq \rho$
(Proposition 3)
Commitment
$C^* = C_\rho^{DRO}(\varepsilon)$
submitted to the
DR operator
radius ε: leave-one-out
validation on Ω^{DA}
the same controller
is back-tested
commitments C_k,
terminal SOC
references
Intraday layer (every 15 min)
Measure state $\mathbf{x}_t$:
backlogs $B_{1..3}$, SOC E_t,
cold S_t, gas engine,
event confirmations
Forecasts corrected by
the latest errors (JR-D)
or scenario tree (JR-S)
Cross-event joint recovery
MILP (2)-(11), 18-h horizon,
all confirmed events:
which debt to repay first
Apply first step; plant
replay (storage, TES,
compute throttling, EDF);
settlement records
$t \leftarrow t + \Delta t$